Kirillov Ch. 2. exercises.

2.1. G Lie group, H closed Lie subgroup, i.e. subgroup + submfd.

1) $\overline{H}$ is subgroup.

2) $\forall x \in \overline{H}$, Hx is open + dense in $\overline{H}$

3) $\overline{H} = H$.

Pf. 1) H subgroup $\Rightarrow e \in H \subset \overline{H}$.

$x, y \in \overline{H} \Rightarrow \forall U \ni x$, $U \cap H = \emptyset$.

$\forall C \supset H$, $x, y \in C$.

closed set is closed under $\cdot$?

Serre Harvard lectures.

$A = M_n(k)$.

$(P_a)_a \subset k[x_1, \dots x_m]$

$G(k) = V(P_a)$

$k' = k[\varepsilon]/\varepsilon^2$.

G finite gp. $H \leq G$ p-Sylow. V f.d. k rep.

$\cup$

W subrep.

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k infinite field, V, W G-reps $/k$. s.t. $V \otimes_k \bar{k} \underset{\bar{k}}{\cong} W \otimes_k \bar{k}$

then $V \underset{k}{\cong} W$

$\rho: Q_8 \longrightarrow \mathbb{H}$ left multiplication. hom ✓

irrep ✓

$\mathrm{End}_G(\rho)$ noncomm.

$\parallel$

$\{ f: \mathbb{H} \to \mathbb{H}$ linear: $f(iv) = if(v)$, $f(jv) = jf(v)$, $f(kv) = k f(v) \}$

$g \neq G$ finite. $\# \mathrm{Conj}\, G = c$. V largest irrep (dim d). Then $\frac{|G|}{c} \leq d^2 \leq |G| - 1$.

① ②

we know $|G| = \sum_{\rho_i \text{ irrep}} \dim \rho_i^2 \leq c \cdot d^2$ ✓ ①

$\#$ irreps $= c$ ②: $|G| - 1 \geq d^2$.

$H \trianglelefteq G$. $G \circlearrowright H$ acts by <u>inner</u> automorphisms. V G-irrep$/\mathbb{C}$. Then $V|_H = W^{\oplus n}$ for H-irrep W.

$g \cdot h = ghg^{-1} = x_g h x_g^{-1}$ for some $x_g \in H$.

$\rho: G \to GL(V)$ $\dim V|_H < \infty \Rightarrow \exists W$ subrep. i.e. H-invariant

$\rightsquigarrow \rho|_H: H \to GL(V)$. $g \in G \rightsquigarrow gW = \{g \cdot w : w \in W\}$ is H-invariant. Why?

$h \cdot gw = \underbrace{gg^{-1}hg}_{\in H \text{ so}} w = g h' w \in gW$.

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$\phi_g : W \to gW$

$w \mapsto g x_g^{-1} w$. linear ✓

H-hom: $\phi_g(hw) = g x_g^{-1} h w$

$= g \underbrace{x_g^{-1} h x_g}\, x_g^{-1} w$

$= g g^{-1} h g x_g^{-1} w$

$= h g x_g^{-1} w$

$= h \phi_g(w)$.

iso: inverse $\phi_{g^{-1}} : gW \to W$,

$V_W = \bigoplus_{W \subset V_H}$

$\psi_g : gW \to W$

$v \mapsto x_g g^{-1} w$

$x_g g^{-1}(hv) = x_g g^{-1} h g g^{-1} v$

$= x_g x_g^{-1} h x_g g^{-1} v$

$= h x_g g^{-1} v$.

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6. G finite gp. not $Z(\mathbb{C}[G]) \cong_{\text{Rings}} \mathbb{C}^r$, $r = \#$ irreps of G/iso

addition: linear — addition: componentwise

mult: $e_g \cdot e_h = e_{gh}$ — mult: complex.

$$\left\{ \sum_{g \in G} \lambda_g e_g \text{ s.t. } \forall h \in G,\ \sum_{g \in G} \lambda_g e_{h^{-1}gh} = \sum_{g \in G} \lambda_g e_g \right\}$$

$\updownarrow$

$\psi \in Cl(G)$ i.e. $G \to k$

$\psi(g) = \psi(h^{-1} g h)$

$\phi(g)$

$\mathbb{C}^r \to \#$irreps $= \#$ conj classes.

$$\rightsquigarrow \left(\sum_{g \in G} \psi(g) e_g\right)\left(\sum_{h \in G} \phi(h) e_h\right)$$

$$= \sum_{g,h} \psi(g) \phi(h)\, e_{gh}$$

$$= \sum_{g \in G} e_g \underbrace{\sum_{xy=g} \psi(x)\phi(y)}_{=(\psi * \phi)(g)}.$$

$$\sum_{g \in G} \lambda_g e_g \rightsquigarrow \psi(g) = \lambda_g \rightsquigarrow \begin{pmatrix} \psi(K_1) \\ \vdots \\ \psi(K_r) \end{pmatrix}.$$

ring iso where $Cl(G)$ convolution

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G finite. $G \underset{\text{trans}}{\circlearrowright} X$ $|X| > 1$.

$\hookrightarrow \forall x, y \in X \ \exists g \in G : gx = y$. $\rightsquigarrow \mathbb{C}^X$ is G rep by

$f : X \to \mathbb{C}$

$\{g$

$g \cdot f : x \longmapsto f(g^{-1} \cdot x)$.

$\mathbb{C}^X$ has a trivial subrep $\omega = \{f : X \to \mathbb{C} : f(x) = f(y) \ \forall x, y \in X\}$.

$\{$

$g \cdot f : x \longmapsto f(g^{-1} x) = f(x)$

so $\rho_\omega(g) = id_{GL(\omega)}$

$V = \mathbb{C}^X / \omega$. <u>Show</u>: irrep $\Longleftrightarrow$ $G \circlearrowright X$ doubly trans, i.e. $\forall x_1, x_2, y_1, y_2 \in X$

$\exists g \in G : gx_1 = x_2,$

$gy_1 = y_2.$

(a) # distinct orbits = # times ρ contains trivial rep

$\rightsquigarrow$ deduce $(\chi | 1) = c$

$\rightsquigarrow G$ trans $\Rightarrow \rho = 1 \oplus \theta$ ✓

(b) $G \circlearrowright X \times X$ by $g(x, y) = (gx, gy)$ $\rightsquigarrow \mathbb{C}^{X \times X}$ has character $\chi_{\mathbb{C}^X}^2$.

(c) TFAE: (i) G doubly trans

(ii) $G \circlearrowright X \times X$ has 2 orbits, Δ and Δ^c.

(iii) $(\chi^2 | 1) = 2$.

(iv) θ is irrep.

(b)

$f : X \times X \to k \overset{g}{\rightsquigarrow} g \cdot f : X \times X \to k$

$(x, y) \longmapsto f(g^{-1}x, g^{-1}y)$.

$\mathbb{C}^{X \times X}$ has basis. $f_{ij}(x_k, x_\ell) = \begin{cases} 1 & i = k, j = \ell \\ 0 & \text{else.} \end{cases}$

$x_k, x_\ell \in X$. $1 \le i, j \le |X|$. i.e $\{f_{ij}\} = |X|^2$.

$\chi_{\mathbb{C}^{X \times X}}(g) = $ # fixed pts of $g : \mathbb{C}^{X \times X} \to \mathbb{C}^{X \times X} =$

$g \cdot f = f \Leftrightarrow \forall x, y : f(x, y) = f(g^{-1}x, g^{-1}y)$

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